

Rigorous Review of The Bell Boundary Condition

Executive summary

The uploaded manuscript is a short conceptual paper submitted to *Symmetry* that presents itself as “Paper V” in a broader brane-lattice program. Its central claim is strong: given a deterministic, nearest-neighbor, time-symmetric substrate, Bell’s theorem allegedly leaves retrocausality as the **unique** consistent completion; the paper then argues that a branching “worldtube” ontology reproduces entangled-photon statistics, no-signalling, the familiar $\cos 2(a - b)$ polarization correlation, Tsirelson saturation, and the Born rule as a form of statistical-mechanical typicality. The manuscript explicitly imports the essential substrate equations and constructions from earlier papers rather than deriving them here, and it lists only a short symbol appendix plus a repository in the data-availability statement.

My overall assessment is that the paper contains a **mixture of standard, correct quantum-mechanical formulas and highly nontrivial, currently unproven interpretive leaps**. The mathematically strongest part is Section 4.5: if one assumes the standard Bell state $|\Phi^+\rangle$, standard qubit polarization observables, and standard Hilbert-space machinery, then the reduced-state no-signalling calculation, the polarization correlation $E(a, b) = \cos 2(a - b)$, and the CHSH value $2\sqrt{2}$ at the canonical settings are all correct. However, none of those results is actually derived from the proposed substrate in this paper; they are *inserted at the level of ordinary quantum formalism* and then redescribed in the language of worldtubes.

The weakest parts are the “uniqueness” argument and the Born-rule discussion. The paper treats Bell’s theorem as if its force were reduced because the underlying substrate is “non-probabilistic at its core,” but Bell-type arguments already apply to deterministic hidden-variable theories; determinism is not an escape hatch, it is one of the cases Bell explicitly targets. The literature also does **not** support the claim that retrocausality is uniquely forced by time-symmetric locality plus determinism. Explicit local deterministic models with relaxed measurement independence already exist in the literature; retrocausal and superdeterministic models have both been studied; and there is active disagreement over whether time symmetry implies retrocausality at all. Moreover, Wood and Spekkens argue that causal explanations of nonsignalling Bell violations, including retrocausal ones without causal cycles, generally require fine-tuning. ¹

Empirically, the paper does not yet meet research-article standards. It presents **no new data, no simulation outputs, no statistical analysis, no ablation study, no quantitative fit to Bell-test data, no version-pinned code audit, and no self-contained derivation of the imported substrate equations**. The 2015 loophole-free Bell experiments and later cosmic Bell tests are highly relevant background, but the manuscript does not quantitatively confront them. The Nobel Prize in Physics 2022 recognized experiments with entangled photons establishing Bell-inequality violation, so any substrate-based paper making comparable explanatory claims must hold itself to a very high standard of formal and empirical specificity. ²

My publication recommendation is therefore: **not publishable in its current form as a physics research article**. It could become publishable only after **major revision**, and even then probably in one of two reframed forms: either as a strictly **conceptual foundations paper** with far more careful claims

and a strong engagement with the retrocausality/superdeterminism literature, or as a **formal/mathematical paper** that actually proves the substrate-to-quantum bridge and supplies reproducible simulations. [filecite turn0file0](#)

Materials reviewed and current verification scope

I reviewed the full uploaded manuscript. It is an 11-page article with a short Appendix A, a references list, and a data-availability statement pointing to a code repository. No separate supplementary PDF, derivation appendix, raw simulation outputs, or code snapshot were supplied in the conversation. The paper itself also states that key ingredients are imported from earlier papers in the series, especially equations from “Paper I” and a localized-mode result from “Paper IV.” [filecite turn0file0](#)

That creates an immediate scope constraint. I can rigorously validate the internal logic and the standard quantum-mechanical algebra that is written down in this manuscript, and I can cross-check its literature framing against the foundations literature. I **cannot** fully validate the claims that depend on missing external artifacts: the substrate action, the hyperbolic PDEs, the claimed nearest-neighbor locality structure, the mapping from substrate variables to a complex two-component carrier, the detector coupling model, the chirality construction, or the numerical behavior of the stated simulator, because those depend on Papers I-IV and repository contents that were not inspectable here. [filecite turn0file0](#)

The exact materials still required for a full end-to-end validation are below.

Required artifact	Why it is necessary	Status in this review
Full texts of Papers I-IV	Needed to verify imported equations (especially the cited Eq. 17, 20, 24, 26), definitions, and the claimed substrate ontology	Missing
Any supplementary derivation notes	Needed to check omitted proofs behind the Bell-state emission, chirality, and Born-typicality claims	Missing
Stable code snapshot or release DOI for BraneSim	Needed for reproducibility, code audit, and reanalysis	Missing in inspectable form
Commit hash matching the manuscript	Needed to ensure the code corresponds to this exact version of the paper	Missing
Run scripts / notebooks / configs	Needed to reproduce any figures or claimed Bell-style simulations	Missing
Raw outputs, logs, parameter files, and environment spec	Needed for numerical replay and stress tests	Missing
Formal statement of the two-time branching boundary-value problem	Needed to test well-posedness, existence, uniqueness, and stability	Missing

Required artifact	Why it is necessary	Status in this review
Detector model details and loss model	Needed to assess fair-sampling, no-signalling under inefficiency, and data realism	Missing

The manuscript's core dependency structure is itself a major review issue:

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flowchart TD
    A[Imported substrate assumptions from Papers I to IV] --> B[Determinism D]
    A --> C[Nearest-neighbor locality L]
    A --> D[Time-symmetric action]
    B --> E[Bell constraint]
    C --> E
    E --> F[Relax MI]
    F --> G[Reject superdeterminism]
    F --> H[Adopt retrocausal worldtube]
    H --> I[Future settings influence past source state]
    I --> J[Assume emitted Bell state  $\Phi^+$ ]
    J --> K[No-signalling calculation]
    J --> L[Correlation  $E = \cos^2(a-b)$ ]
    J --> M[Tsirelson saturation]
    J --> N[Born weight from Liouville typicality]

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As a logic map, this is easy to follow. As a proof map, however, nodes **G**, **J**, and **N** are not demonstrated in the present paper. [filecite turn0file0](#)

Step-by-step critique of the argument

Bell-theorem framing

The paper correctly identifies the usual Bell-type triad it wants to discuss—determinism, locality, and measurement independence—and it is right that modern Bell experiments have strongly constrained local realist explanations. The cited 2015 loophole-free Bell tests closed the major locality and detection loopholes, and later cosmic-setting Bell tests pushed any common-cause explanation much deeper into the past light cone. That background is solid. [filecite turn0file0](#) 3

Where the paper begins to overreach is in Section 3.3, where it suggests Bell's theorem presupposes that the fundamental description of a single run is "irreducibly statistical," and that the deterministic substrate therefore departs from the "probabilistic premise." That is misleading. Bell inequalities are constraints on ensemble statistics, but **deterministic hidden-variable models are already within Bell's target class**; in fact, Bell's original framing and later literature treat deterministic response functions as a special case of the general probabilistic framework. Hall explicitly discusses local deterministic models obtained by relaxing measurement independence, and Wharton–Argaman discuss locally mediated models that preserve locality only by relaxing an arrow-of-time assumption. So the manuscript is right that Bell does not go away, but wrong to present its deterministic substrate as standing outside the theorem's natural domain. [filecite turn0file0](#) 4

This matters because the paper’s rhetoric understates the burden of proof. A deterministic substrate does **not** reduce the need for a rigorous derivation of the observable probability law; it increases it. Once the paper claims to recover Bell-test statistics, it must show exactly how deterministic histories induce operational probabilities over runs, settings, and outcomes. The manuscript explicitly admits that this derivation remains open. [filecite turn0file0](#)

The uniqueness claim

The paper’s strongest rhetorical claim is that, for *this* substrate, retrocausality is not merely available but **uniquely forced**. That is not established. The manuscript offers a philosophical preference ordering, not a theorem. It rejects superdeterminism because the substrate is “not a pre-programmed Oracle,” and because explanatory fine-tuning is said to be unattractive; but explicit local deterministic measurement-dependent models do exist, including Brans-style and Hall-style constructions, and Hall & Branciard later quantified retrocausal and causal measurement-dependence costs in the CHSH scenario. Those works do not prove superdeterminism true, but they do show that “measurement dependence” is a wider design space than the paper acknowledges. [filecite turn0file0](#) 5

The paper also argues that retrocausal/all-at-once theories place the correlation “in the law” rather than “in tuned past data.” That line is compatible with part of the Lagrangian-schema literature, but it does not by itself yield uniqueness. Wharton and Argaman present locally mediated reformulations as an important research program, not as a settled uniqueness theorem. Worse for the manuscript, Wood and Spekkens argue that causal explanations of Bell-inequality violations subject to no-signalling typically require fine-tuning of causal parameters, and they explicitly include retrocausal influences among the cases that face this challenge. That directly undercuts the paper’s suggestion that retrocausality here is uniquely non-conspiratorial. 6

A rigorous uniqueness claim would need a formal theorem of the following kind: given the exact substrate state space, dynamics, boundary conditions, admissible observables, and measurement-setting variables, every Bell-violating completion compatible with the substrate either reduces to retrocausality or violates a stated axiom. No such theorem appears in the manuscript.

[filecite turn0file0](#)

Time symmetry, arrow of time, and chirality

The manuscript’s distinction between microscopic time symmetry and a thermodynamic arrow is reasonable as a conceptual stance. Time-symmetric dynamical laws plus asymmetric boundary conditions are standard in statistical mechanics and in some retrocausal foundations programs.

[filecite turn0file0](#) 7

However, the paper then turns that stance into a stronger conclusion than the literature supports. The question whether time symmetry implies retrocausality is **debated**. Leifer and Pusey argue that under a precise set of assumptions a non-retrocausal time-symmetric ontology is impossible; Maudlin disputes that argument and claims the proof contains a fatal error. The manuscript cites only the pro-retrocausal side and presents the matter as essentially settled. That is not an adequate literature review for a paper making a uniqueness claim. 8

The chirality discussion also overclaims relative to the paper’s own admissions. The manuscript uses chirality to explain the matter/antimatter distinction and even a cosmological matter–antimatter asymmetry, but the open-problems section later says the chirality quantum number still needs to be defined precisely and shown to reverse under $t \mapsto -t$. That is a serious internal tension: a mechanism

cannot count as an explanation at research-paper level when its key state variable is still declared undefined. [filecite turn0file0](#)

Entangled-photon section and physical interpretation

The paper’s discussion of the laboratory Bell setup mixes correct standard quantum-optical background with an underjustified substrate reinterpretation. It is true that Bell tests with entangled photons are central to the experimental story recognized by the 2022 Nobel Prize, and it is sensible to use polarization-entangled photons as the benchmark phenomenon. [9](#)

But the manuscript moves too quickly from that benchmark to its own source model. It cites type-II down-conversion experiments, then declares that the substrate’s natural output is the symmetric Bell state $|\Phi^+\rangle$, while describing that as “the type-I-like symmetric output.” In standard SPDC language, type-I and type-II refer to different polarization structures, and type-II sources characteristically produce orthogonally polarized daughter photons before compensation/interferometric engineering. The paper may still be able to map the laboratory state to $|\Phi^+\rangle$ up to local unitaries, but that mapping needs to be written out explicitly rather than asserted. [filecite turn0file0](#) [10](#)

The statement that photon randomness “actually lives” only at the detector, because the radiation in flight is a definite continuous carrier while quantization belongs solely to bound matter solitons, is also much too strong as written. Mainstream quantum optics treats the electromagnetic field itself as quantized; fully quantum models of light–matter interaction are a core part of the field. The manuscript can propose an alternative ontology, but it cannot present that ontology as if it were already an established consequence of the substrate, especially without a derivation of the observed nonclassical optical signatures. [filecite turn0file0](#) [11](#)

Reproduced derivations and mathematical validation

Verification summary

The key mathematical claims can be separated cleanly into two groups: standard quantum-mechanical identities that are correct, and substrate-specific bridges that are not derived.

Item in manuscript	Validation outcome	Assessment
Reduced-state no-signalling formula	Correct , conditional on standard QM assumptions	Mathematically sound
Polarization correlation $E(a,b) = \cos 2(a - b)$	Correct , conditional on Bell state and chosen observables	Mathematically sound
CHSH saturation at the canonical angles	Correct	Numerically and symbolically sound
Emission of $ \Phi^+\rangle$ from the substrate’s three-wave vertex	Not established here	Missing derivation from imported equations
Born rule from Liouville/symplectic typicality	Not established	Heuristic at best
Uniqueness of retrocausal completion	Not established	Philosophical argument, not theorem

Reproduction of the no-signalling step

The paper writes the Bell state as

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|H\rangle_A |H\rangle_B + |V\rangle_A |V\rangle_B).$$

Starting from the projector,

$$|\Phi^+\rangle\langle\Phi^+| = \frac{1}{2} (|HH\rangle\langle HH| + |HH\rangle\langle VV| + |VV\rangle\langle HH| + |VV\rangle\langle VV|),$$

the partial trace over Alice yields

$$\rho_B = \text{Tr}_A(|\Phi^+\rangle\langle\Phi^+|) = \frac{1}{2} (|H\rangle\langle H| + |V\rangle\langle V|) = \frac{1}{2} I.$$

Therefore for any normalized Bob basis vector $|b_B\rangle$,

$$P(B | a, b) = \langle b_B | \rho_B | b_B \rangle = \frac{1}{2}.$$

So Equation (2) is correct as ordinary quantum mechanics. The paper is also right that, at this formal level, the distant setting does not appear in Bob's marginal. filecite turn0file0

The problem is not the algebra; the problem is the **claim of derivation**. In the manuscript, $\rho_B = I/2$ follows only after the entangled state and the Hilbert-space formalism have already been assumed. The substrate does not yet deliver the reduced density operator from its own microdynamics. So the result is presently a **restatement of standard QM inside the proposed ontology**, not a substrate-level theorem. filecite turn0file0

There is also an important experimental nuance. The paper says this no-signalling step uses “only completeness of the local measurement — that every photon is detected (unit efficiency / fair sampling).” That is too compressed. The standard no-signalling theorem is a statement about complete local quantum operations; in loophole-free Bell tests, fair-sampling assumptions are precisely what one tries to avoid, and the 2015 photon experiments were designed so that Bell violation did not rely on them. Realistic loss can be handled, but only if the loss model and the operational sample space are specified carefully. filecite turn0file0 12

Reproduction of the correlation law and Tsirelson saturation

The manuscript defines the analyzer observable as

$$A_a = s(a) \cdot \sigma, \quad s(a) = (\sin 2a, 0, \cos 2a),$$

with the same form for Bob at angle b . For the Bell state $|\Phi^+\rangle$, the correlation matrix in the (x, y, z) basis is effectively $T = \text{diag}(1, -1, 1)$, and because the chosen measurement vectors live in the x - z plane, one gets

$$E(a, b) = \langle \Phi^+ | A_a \otimes B_b | \Phi^+ \rangle = \sin 2a \sin 2b + \cos 2a \cos 2b = \cos 2(a - b).$$

So Equation (3) is correct. [filecite turn0file0](#)

At the cited CHSH settings $(a, a', b, b') = (0, \pi/4, \pi/8, 3\pi/8)$, the four correlations are

$$\begin{aligned} E(a, b) &= \cos(-\pi/4) = \frac{\sqrt{2}}{2}, & E(a, b') &= \cos(-3\pi/4) = -\frac{\sqrt{2}}{2}, \\ E(a', b) &= \cos(\pi/4) = \frac{\sqrt{2}}{2}, & E(a', b') &= \cos(-\pi/4) = \frac{\sqrt{2}}{2}. \end{aligned}$$

Hence

$$|S| = |E(a, b) - E(a, b') + E(a', b) + E(a', b')| = \left| \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2} \right) + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right| = 2\sqrt{2}.$$

So Equation (4) is also correct. The algebra checks out. [filecite turn0file0](#)

Again, the weakness is the bridge. The paper says these results follow because the substrate carrier is $\mathbb{C}^2$ and therefore admits genuine Hilbert overlaps capped by Tsirelson's theorem. That may be plausible *if* Papers I–III truly derive the exact qubit Hilbert structure and measurement operators, but those derivations are not provided here. In the current paper, the substrate-to-qubit map is assumed rather than proved. [filecite turn0file0](#) ¹³

Counterexample-style critique of the Born-weight argument

The Born-rule section is the least convincing mathematical part of the paper. The manuscript argues that because a complex amplitude can be written $\psi = q + ip$, it is a two-dimensional symplectic phase space, and the equal-a-priori Liouville measure on that space makes $|\psi|^2$ the probability because the exponent 2 is “the symplectic dimension.” The paper itself later admits that establishing the equilibrium measure is a “residual probabilistic debt,” which already signals that the argument is not complete.

[filecite turn0file0](#)

A simple counterexample reveals the gap. Suppose a two-outcome experiment has amplitudes (α, β) with $|\alpha|^2 + |\beta|^2 = 1$. The Born rule says $P(A) = |\alpha|^2$ and $P(B) = |\beta|^2$. But a Liouville measure on the *individual* complex plane for one amplitude does not, by itself, define a probability over the *pair* of alternatives. One still needs at least:

- the correct joint state space for all amplitudes together,
- a normalization rule,
- a measure on that normalized space,
- and a map from microstates to macroscopic outcomes.

Without those ingredients, “ $|\psi|^2$ is an area” is only a dimensional observation, not a derivation of outcome frequencies. The argument may hint at *why* a square norm is natural, but it does not derive the Born rule in the operational sense used in Bell experiments. [filecite turn0file0](#)

There is another internal inconsistency: Section 4.5 presents the Born weight as if the core issue were solved, whereas Section 4.7 explicitly says that relaxation to the equilibrium measure remains an open problem and calls it the remaining “probabilistic debt.” Those two claims cannot both be true in a strong sense. Either the Born rule has been derived, or it remains conjectural. The paper must choose one formulation and state it consistently. [filecite turn0file0](#)

Reproducibility, empirical adequacy, and replication experiments

Reproducibility checklist

In its current form, the paper scores poorly on reproducibility as a research article, even allowing for its conceptual emphasis.

Reproducibility item	Status	Review comment
Main manuscript available	Yes	Reviewed in full. filecite turns file
Prior papers required for core equations	No	Essential dependencies missing from the review package. filecite turns file
Supplementary derivations	No	None provided here.
Public code location mentioned	Yes	Repository is named in the paper, but not inspectable here as an auditable artifact. filecite turns file
Pinned version / commit hash	No	Missing.
Simulation parameters / hyperparameters	No	Missing.
Raw outputs and logs	No	Missing.
Statistical analysis plan	No	Missing.
Comparison against published Bell-test counts/correlations	No	Missing.
Detector inefficiency model	No	Missing.
Formal existence/uniqueness proof for two-time BVP	No	Missing; paper lists it as open. filecite turns file

The practical implication is clear: the manuscript is not yet reproducible enough to support claims of physical explanation, especially claims that it reproduces Bell-test statistics in a novel underlying ontology. [filecite turns file](#)

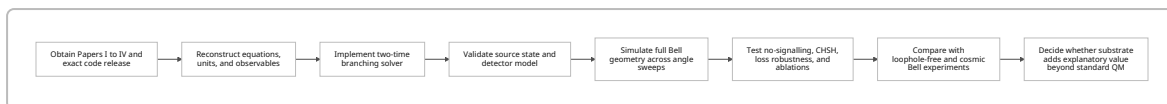
Suggested replication experiments

The most useful way forward is not more rhetoric but a sequence of replication-grade tests. These should be pre-specified and version-controlled.

Replication experiment	Goal	Minimal required output	Success criterion
Two-time Bell-boundary solver	Solve the actual branching boundary-value problem with future settings as boundary inputs	Existence, convergence diagnostics, uniqueness/stability study	A well-posed solver that converges across settings and initial conditions

Replication experiment	Goal	Minimal required output	Success criterion
Substrate-to-qubit bridge test	Show how a substrate state maps to a normalized $\mathbb{C}^2$ polarization state and local analyzer observable	Explicit map, operators, invariants, unit tests	Recovery of standard polarization probabilities without ad hoc insertion
Bell-correlation replay	Reproduce $E(a, b)$ over a sweep of analyzer angles	Angle-resolved correlation curve and residuals vs $\cos 2(a - b)$	Quantitative agreement within stated tolerance
CHSH benchmark	Compute S on the canonical settings and under perturbed settings	Point estimate and uncertainty procedure	Robust approach to $2\sqrt{2}$ in the ideal limit, with deviations under controlled nonidealities
Loss and detection-ablation study	Test whether no-signalling and Bell violation survive realistic inefficiencies	Simulations with explicit loss channels, sample-selection rules, and detection model	Demonstration that conclusions are not hidden fair-sampling artifacts
"No future boundary" ablation	Test whether retrocausal input is truly doing the explanatory work	Matched simulation with future-setting dependence disabled	CHSH falls back to the local bound or another clearly weaker regime
Chirality-pair simulation	Validate particle/antiparticle identification claim	Paired solutions, conserved quantities, time-reversal relation	Explicit paired solutions with mathematically defined chirality
Data-level comparison to experiments	Compare model outputs against published Bell tests	Fit to reported correlations, visibilities, and p-values where possible	Predictive or at least competitive explanatory adequacy against experiment

These experiments should be linked in a transparent workflow:



Without this program, the manuscript remains a speculative interpretation rather than a validated theoretical result. [filecite](#) [turn0file0](#)

Publication recommendation and target journals

Recommendation

My recommendation is **reject in current form**, with an invitation to resubmit only after major reconstruction. The reason is not that the paper is unintelligent or unreadable. On the contrary, it is

readable and ambitious. The problem is that it currently asks the reader to accept too many decisive steps without theorem-level proof or replication-level evidence. [filecite turn0file0](#)

If revised successfully, the best publication strategy depends on how the manuscript is reframed. If the author supplies formal derivations and code-backed evidence, it could become a quantum-foundations or mathematical-physics submission. If instead it remains primarily interpretive, it should be recast as a conceptual/philosophical foundations paper with much more modest claims. The table below compares the most plausible venues.

Journal	Official scope summary	Recent impact factor	Likely audience	Fit for this manuscript
Foundations of Physics	Forum for conceptual structures and foundational aspects across relativity, quantum theory, field theory, thermodynamics, cosmology, and information theory	1.0 (2024)	Physicists, mathematicians, philosophers working on foundations	Best fit if heavily revised into a careful foundations paper with toned-down claims and formal support. ¹⁴
Quantum Studies: Mathematics and Foundations	Deeper understanding of fundamental aspects of quantum theory, linking theoretical questions, foundational issues, and mathematical methods	1.0 (2024)	Quantum-foundations researchers, mathematical physicists, philosophers of science	Good fit only if the substrate-to-quantum bridge is made mathematically explicit and reproducible. ¹⁵
Studies in History and Philosophy of Science	Integrated study of history, philosophy, and sociology of science; includes modern-physics/philosophy material after merger of the former modern-physics journal line	1.8 Impact Factor; 3.2 CiteScore	Historians and philosophers of science, foundations scholars	Best fit if reframed as a conceptual intervention on Bell, retrocausality, and explanatory standards rather than as a solved physical model. ¹⁶

I would **not** recommend targeting a higher-barrier mainstream physics journal at this stage. Before that, the author would need a self-contained formal model, explicit proofs where strong claims are made, and a reproducible computational or empirical package. [filecite turn0file0](#)

Prioritized revisions

The most important revisions are substantive, not cosmetic.

Priority	Revision	Why it matters most
Critical	Supply a self-contained theorem or derivation for the substrate-to-qubit bridge, including the source state, analyzer operators, and detector model	Without this, the core Bell results are standard QM formulas pasted onto the substrate, not substrate predictions
Critical	Remove or greatly weaken the claim that retrocausality is the unique consistent completion unless a formal uniqueness theorem is proved	Current argument is philosophical and incomplete, and the literature does not support uniqueness as stated
Critical	Rewrite the Born-rule section so that it is either a rigorous derivation or clearly labelled a conjectural heuristic	The current presentation overstates what has been shown
Critical	Provide Papers I–IV, exact equation references, and a reproducible code package with pinned version and run scripts	Core claims are currently unverifiable
High	Engage the omitted literature: Brans/Hall on measurement dependence, Hall & Branciard on causal vs retrocausal costs, Wood & Spekkens on fine-tuning, Leifer–Pusey and Maudlin on time symmetry and retrocausality, and later Bell-test literature on freedom-of-choice style constraints	Needed for a fair and current foundations review. 17
High	Resolve terminology: distinguish Bell locality, signal locality, nearest-neighbor dynamical locality, measurement independence, free settings, contextuality, and all-at-once constraint satisfaction	Several of these notions are blurred together in the present draft
High	Add quantitative benchmarking against actual Bell data, not just qualitative alignment with the familiar formulas	A physical model should be tested against experiments already recognized as foundational landmarks. 9
Medium	Clarify the SPDC source discussion and explicitly map any laboratory source state to the chosen Bell state, including local-unitary conventions	Needed to avoid confusion between the cited experimental apparatus and the paper’s chosen state
Medium	Do not use chirality to explain matter/antimatter asymmetry until chirality is mathematically defined and demonstrated in solutions	The paper currently relies on an undeclared quantity that it later admits is still open

Priority	Revision	Why it matters most
Medium	Separate “what is proved” from “what is interpretive motivation” in every section	This will significantly improve clarity and scholarly trust

On balance, the manuscript has the outline of an interesting foundations paper, but not yet the evidentiary or formal standard of a publishable research article. Its best path forward is to become **narrower, more explicit, less absolute, and far more reproducible.** turn0file0

    The significance of measurement independence for Bell inequalities and locality
https://arxiv.org/abs/1511.00729?utm_source=chatgpt.com

   Significant-loop-hole-free test of Bell's theorem with entangled photons
https://arxiv.org/abs/1511.03190?utm_source=chatgpt.com

 Bell's Theorem and Locally-Mediated Reformulations of Quantum Mechanics
https://arxiv.org/abs/1906.04313?utm_source=chatgpt.com

 Does Time-Symmetry Imply Retrocausality? How the Quantum World Says "Maybe"
https://arxiv.org/abs/1002.0906?utm_source=chatgpt.com

 Is a time symmetric interpretation of quantum theory possible without retrocausality?
https://arxiv.org/abs/1607.07871?utm_source=chatgpt.com

 The Nobel Prize in Physics 2022 - NobelPrize.org
<https://www.nobelprize.org/prizes/physics/2022/summary/>

 Spontaneous parametric down-conversion
https://en.wikipedia.org/wiki/Spontaneous_parametric_down-conversion?utm_source=chatgpt.com

 Jaynes–Cummings model
https://en.wikipedia.org/wiki/Jaynes%E2%80%93Cummings_model?utm_source=chatgpt.com

 Tsirelson's bound
https://en.wikipedia.org/wiki/Tsirelson%27s_bound?utm_source=chatgpt.com

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<https://www.sciencedirect.com/journal/studies-in-history-and-philosophy-of-science>